SIMATS ENGINEERING

ASSESSMENT TOOL 2: Error Analysis Task

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1516

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**

Question Count: 5

Code of Conduct

I (**AMARA SATWIKA / 192525204)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **A.SATWIKA**

Assessment Tasks

Error Analysis Task

1. A Hadoop job repeatedly fails because input splits are uneven and reducers remain idle. Identify the likely design error and propose corrective actions.

Uneven Input Splits and Idle Reducers.

Answer

The main design error is improper input split and partitioning of data. When input splits have highly different sizes, some Mapper tasks take much longer than others. As a result, some reducers remain idle while waiting for mapper outputs.

Problems

- Uneven input file sizes.
- Poor input split configuration.
- Data skew between partitions.
- Improper number of reducers.
- Large mapper tasks delay the shuffle phase.

Corrective Actions

- Use an appropriate HDFS block size.
- Ensure input files are reasonably balanced.

- Use a suitable number of Mapper and Reducer tasks.
- Apply a proper partitioner to distribute keys evenly.
- Use a Combiner to reduce intermediate data.
- Monitor task execution and identify data skew.

2. An HDFS deployment shows frequent data unavailability after a single node failure. Analyze the probable replication or NameNode design error.

HDFS Data Unavailability After Node Failure

Answer

The probable error is insufficient data replication or a single-point-of-failure NameNode design.

HDFS normally stores multiple copies of each block. If only one copy exists on the failed node, the data becomes unavailable.

Probable Causes

- Replication factor set too low.
- Data replicas stored on the same node/rack.
- NameNode is not configured for High Availability.
- Improper replica placement.
- Failure detection or recovery is not properly configured.

Corrective Actions

- Set an appropriate **replication factor, preferably 3**.
- Distribute replicas across different DataNodes.
- Use **NameNode High Availability** with Active and Standby NameNodes.
- Enable automatic failover.
- Monitor DataNode health.
- Allow HDFS to automatically recreate missing replicas.

3. A data ingestion workflow using ETL and Apache NiFi creates duplicate records in the analytics store. Identify the likely causes and recommend fixes.

Duplicate Records in ETL and Apache NiFi

Answer

The main problem is duplicate data ingestion without proper record identification and processing control.

Probable Causes :

- Same source data is processed multiple times.
- NiFi flow is restarted without tracking processed records.
- No unique record ID.
- ETL process performs full extraction repeatedly.
- Database insertion does not check for existing records.
- Improper retry configuration.

Corrective Actions :

- Assign a unique ID to each record.
- Use incremental data extraction instead of full extraction.
- Maintain processing/checkpoint information.
- Use NiFi attributes to track processed data.
- Apply deduplication before storing data.
- Use database constraints such as unique keys.
- Configure retries carefully to avoid duplicate insertion.

Correct flow :

```
Source
  |
NiFi
  |
Unique ID Check
  |
Deduplication
  |
ETL Processing
  |
Analytics Store
```

4. A YARN cluster suffers from resource starvation when multiple users submit jobs together. Analyze the scheduling error and suggest a better resource management approach.

YARN Resource Starvation

Answer

The likely scheduling error is improper resource allocation and lack of queue-based scheduling. One user's large jobs may consume most of the available cluster resources, leaving insufficient resources for other users.

Problems

- No resource limits.
- Poor scheduler configuration.
- One user consumes excessive CPU/memory.
- No separate queues.
- No priority or fairness policy.

Corrective Actions :

- Configure YARN Capacity Scheduler or Fair Scheduler.
- Create separate queues for different users or workloads.

- Set minimum and maximum resource limits.
- Assign appropriate priorities.
- Prevent a single job from consuming all resources.
- Monitor CPU and memory usage.

5. A backup strategy for distributed storage restores outdated data after recovery. Identify the probable design flaw in the replication or backup approach.

Outdated Data After Backup Recovery.

Answer

The probable design flaw is using outdated or improperly synchronized backup copies. The backup system may not be capturing the latest changes before a failure occurs.

Probable Causes

- Backups are performed infrequently.
- Replicas are not synchronized.
- Backup contains old snapshots.
- No incremental backup mechanism.
- Recovery uses an older backup version.
- Backup metadata is not updated properly.

Corrective Actions

- Perform regular backups.
- Use incremental backups to capture recent changes.
- Maintain timestamped backup versions.
- Verify backup consistency.
- Synchronize replicas regularly.
- Use the latest valid snapshot during recovery.
- Test restoration periodically.

Recovery Process :

Latest Full Backup
|
Latest Incremental Backup
|
Reconstruct Data
|
Current Dataset

Error Analysis Task Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Identification of Error	30%	Accurately identifies the root technical issue	Identifies most of the issue	Partially identifies issue	Fails to identify issue
Cause Analysis	20%	Explains root cause with strong technical reasoning	Reasonable cause analysis	Basic cause analysis	Weak analysis
Corrective Action	25%	Proposes effective and feasible corrections	Reasonable corrections	Basic corrections	Ineffective corrections
Use of Hadoop / Integration Concepts	15%	Applies relevant concepts appropriately	Mostly appropriate application	Partial application	Incorrect application
Clarity	10%	Clear and direct explanation	Generally clear	Somewhat unclear	Poorly presented